

Aspen Aerogels, Inc. NYSE:ASPN

FQ3 2021 Earnings Call Transcripts

Thursday, October 28, 2021 9:00 PM GMT

S&P Global Market Intelligence Estimates

	-FQ3 2021-			-FQ4 2021-	-FY 2021-	-FY 2022-
	CONSENSUS	ACTUAL	SURPRISE	CONSENSUS	CONSENSUS	CONSENSUS
EPS Normalized	(0.26)	(0.24)	NM	(0.26)	(0.97)	(0.76)
Revenue (mm)	27.63	30.38	▲9.95	28.85	116.22	141.35

Currency: USD

Consensus as of Oct-25-2021 12:29 PM GMT

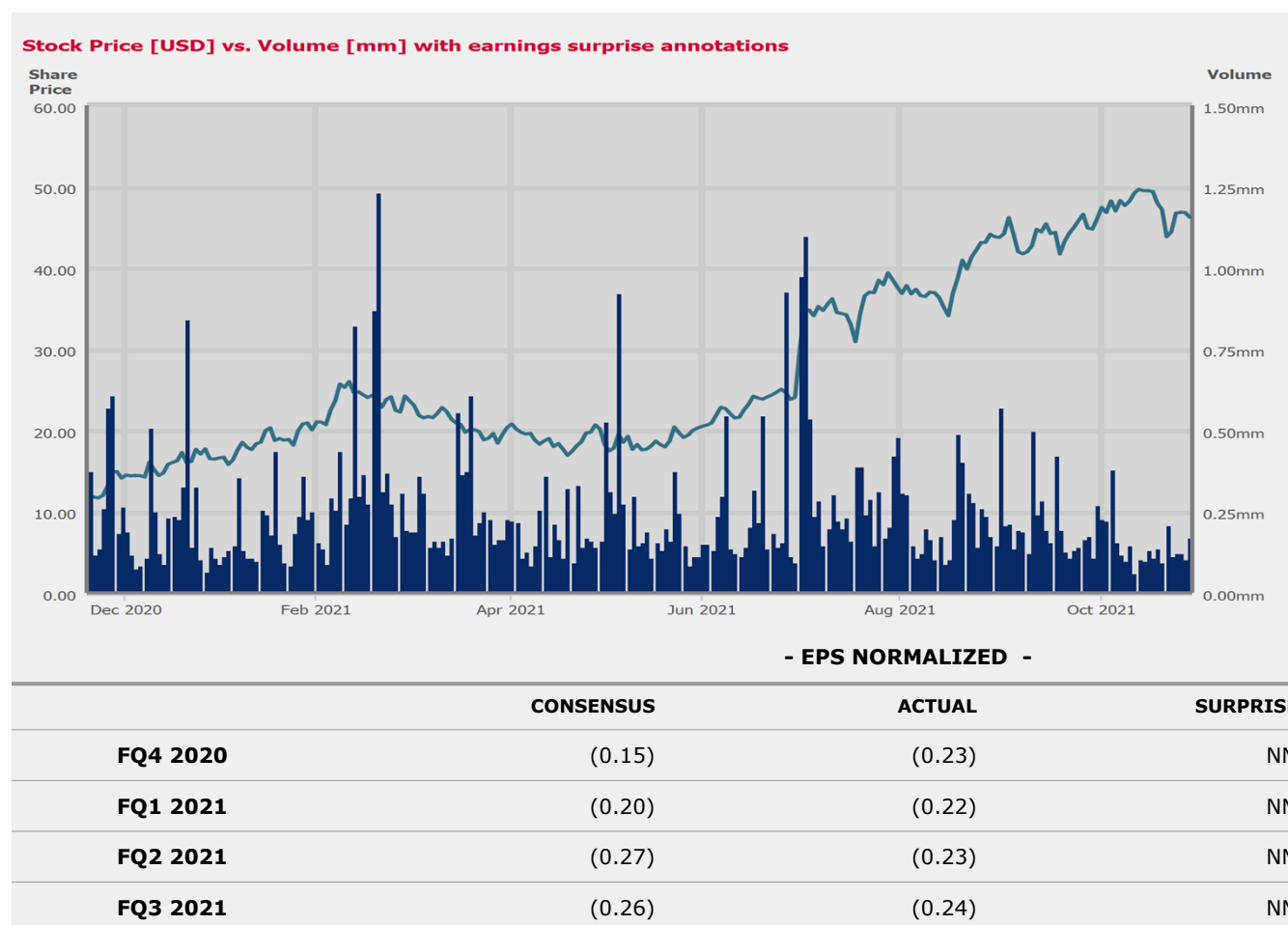


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Call Participants

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Presentation

Operator

Welcome, and thank you for attending the Aspen Aerogels' Q3 2021 Earnings Call. [Operator Instructions] I would now like to pass the conference over to your host, Laura Guerrant, with Aspen Aerogels. Thank you. You may proceed, Ms. Guerrant.

Laura Guerrant-Oiye

Thank you, Sarah. Good evening, and thank you for joining us for the Aspen Aerogels conference call. I'm Laura Guerrant, Aspen's Vice President of Investor Relations and Corporate Communications. There are a few housekeeping items that I would like to address before turning the call over to Don Young, Aspen's President and CEO.

The press release announcing Aspen's financial results and business developments as well as a reconciliation of management's views of non-GAAP financial measures compared to the most applicable GAAP measures is available on the Investors section of the Aspen website, www.aerogel.com. In the press release is a summary statement of operations, a summary balance sheet and a summary of key financial and operating statistics for the third quarter and 9 months ended September 30, 2021. In addition, the Investors section of Aspen's website will contain an archived version of this webcast for approximately 1 year.

Please note that our discussion today will include forward-looking statements, including any statement regarding outlook, expectations, beliefs, projections, estimates, targets, prospects, business plans and any other statement that is not a historical fact. These forward-looking statements are subject to risks and uncertainties. Aspen Aerogel's actual results may differ materially from those expressed in these forward-looking statements. A list of factors that could affect the company's actual results can be found in Aspen's press release issued today and are discussed in more detail on the reports Aspen files with the SEC, particularly in the company's most recent annual report on Form 10-K.

The company's press release issued today and filings with the SEC can also be found in the Investors section of Aspen's website. Forward-looking statements made today represent the company's views as of today, October 28, 2021. Aspen Aerogels disclaims any obligation to update these forward-looking statements to reflect future events or circumstances.

During this call, we will refer to non-GAAP financial measures, including adjusted EBITDA. These financial measures are not prepared in accordance with U.S. generally accepted accounting principles or GAAP. These non-GAAP financial measures are not intended to be considered in isolation or as a substitute for results prepared in accordance with GAAP. The definitions and reconciliations of these non-GAAP financial measures to the most directly comparable GAAP financial measures and a discussion of why we present these non-GAAP financial measures are included in today's press release.

I'll now turn the call over to Don. Don?

Donald R. Young

President, CEO & Director

Thanks, Laura. Good afternoon, everyone. Thank you for joining us for our Q3 2021 earnings call. I should note that Laura Guerrant joined us a couple of months ago as our VP, Investor Relations and Corporate Communications. She has 30 years of professional experience and is joining us just in the nick of time. John and I have been moonlighting in the role since we went public in 2014, and we welcome Laura as we broaden our outreach to the investment community.

Today, I will describe the key highlights of our progress towards achieving our near-term and longer-term business goals. John will then recap our Q3 performance and finish with an updated outlook for 2021. We will conclude today's call with a Q&A session.

The key points I intend to cover are: the drivers for both strong Q3 revenue and our full year outlook, which we have now raised for the third time this year; the acceleration and expansion of our PyroThin thermal barrier business, which we believe sets the stage for 2022 PyroThin revenue to be 2 or more times our original 2022 revenue estimate; the progress on key investments related to Plant 2; and the technical and business development for Aspen battery materials.

The third quarter was strong with top line growth exceeding our expectations, driven by both our energy infrastructure business and our emerging PyroThin thermal barrier shipments. Revenue grew 26% year-over-year to \$30.4 million, and gross profit increased 61%, reflecting the leverage associated with increasing utilization of our manufacturing plant.

The recovery of our energy infrastructure business is in line with our expectations as we come out of what we hope to be the worst of the COVID pandemic. We predicted that as the impact of the pandemic subsided, energy infrastructure revenue would begin to increase towards pre-pandemic levels as pent-up demand materialized. This has, in fact, been the case as we are seeing high levels of maintenance-related activity especially in key countries with higher vaccination rates.

This positive indicator is an important factor contributing to our 2021 guidance, which we have raised for the third time this year. As a reminder, one of the key components of our goal to double revenue from 2021 to 2023 is to recapture our pre-pandemic energy infrastructure revenue, which averaged nearly \$35 million per quarter during 2019. We believe that we are well positioned to achieve this goal no later than 2023.

Over the longer term, we believe the value of our energy infrastructure products, which is based on resource efficiency, asset resiliency and safety, particularly fire safety, will become yet more important to our customers in the years to come as they strive to attain their own sustainability goals. And when you consider the potential of leveraging our aerogel technology platform into new markets such as hydrogen infrastructure and carbon capture, we see significant growth opportunities over this decade for our energy infrastructure business.

In my earlier comments, I mentioned our emerging shipments of PyroThin thermal barriers. This business is advancing at a faster pace than we initially anticipated. John and I have previously said that we expect 2021 PyroThin thermal barrier revenue to be in the low single-digit millions of dollars, in 2022 PyroThin revenue to be in the mid- to upper single-digit millions.

During Q3, we exceeded \$1 million of PyroThin revenue on a year-to-date basis, and we now expect Q4 revenue alone to exceed \$5 million. This acceleration of revenue, we believe sets the stage for 2022 PyroThin thermal barrier revenue to be 2x or more our initial expectations of mid- to high single-digit millions.

On our last earnings call, we said that we were on the cusp of winning a contract with a major Asia-based automotive OEM with a leading global platform. We have, in fact, received our first orders for that OEM's first all-electric vehicle that will be offered in Asian, European and North American markets. We are delivering the first production parts to this customer in Q4, and it should be noted that this OEM has partnered with another large agent EV OEM that will use the same battery modules to power its electric offering.

We are also delivering production parts to our major U.S. automotive OEM during the fourth quarter. These first deliveries are preloading the pipeline to these customers and enhancing our Q4 revenue. The transition from supplying prototype parts to production parts is a major milestone and signifies thermal barrier parts destined for Q4 and Q1 commercial EV launches. The milestone also represents a prerequisite for announcing the names of these 2 first customers, which we hope to do soon.

There are several drivers accelerating PyroThin thermal barrier revenue, including the transition to production parts with our 2 existing EV OEMs, very active prototyping with new EV OEMs in our business development funnel and cases where we are seeing an increasing scope of work as we establish ourselves as a technology partner to these companies. I discussed earlier the importance of the move to production

parts, and I will discuss the business development funnel in a moment. But first, I want to explain in more depth the importance of Aspen being viewed as a technology partner.

Our approach is threefold. Our intent is for Aspen to be the industry expert in battery thermal management, for PyroThin thermal barriers to be the industry standard and for Aspen to be OEM agnostic. As a technology partner to EV OEMs, we are being asked to focus more broadly on battery performance and safety. This expanded mandate is in addition to our current mission-critical role of mitigating thermal runaway.

Our team is designing more sophisticated aerogel-based solutions to optimize thermal management and mechanical performance during a standard operation of the battery system. Again, this mandate focuses more broadly on battery performance and safety, and the likely result of this advancement is more technical scope and more content per vehicle than we initially estimated. This development is resulting in increased thermal barrier revenue estimates in 2021 and 2022 and very likely in subsequent years.

We've made considerable progress during the quarter in the business development funnel for our PyroThin thermal barriers. The pace of engagement within our 3-stage business development process remains high, and we are responding to multiple formal requests, including RFIs, RFQs and performance validations.

We continue to work with several of the remaining EV OEMs on developing solutions for improved battery safety in the event of thermal runaway as well as for thermal and mechanical management solutions for improved battery performance during the normal course of the batteries operation. With the increased intensity of the EV megatrend and our broader mandate around battery performance and safety, the pace of our commercial work is accelerating and the size of the opportunity is expanding.

In order to aggressively capitalize on our expanding demand profile, we are also expediting the work on our second aerogel manufacturing plant. We have selected a 90-acre parcel in the Southeastern United States as the site of the second plant and are working closely with state and local officials to finalize zoning approvals and incentive arrangements. Subject to the approval of our Board, we anticipate making a joint public announcement of our site selection with the Governor's office in the near future. This site gives us the space to design Plant 2 to provide more than twice the capacity of Plant 1. We plan to have Phase 1 of Plant 2 operational during the second half of 2023.

We are currently working to match the size of the first phase of the plant construction with the accelerated and expanded demand profile described earlier. We are also keenly focused on building a plant with low operating costs and one that has key raw materials generated onsite or nearby.

On a related note, it has been about 4 months since Koch Strategic Platforms made a \$75 million investment in Aspen. The premise of the investment was not only to strengthen our balance sheet, but also to bring to bear the broader resources of the significant Koch network to enable us to scale more effectively. Plant 2 engineering is a good example.

We have engaged Koch project solutions to support our own outstanding strategic capital projects team with additional resources drawn from its vast experience. We expect this support will continue for the duration of the project, which includes Koch providing onsite project leadership during the construction phase.

There are additional opportunities for us to work with teams from Koch in areas such as the fabrication of thermal barrier parts in Mexico and the optimization of purchasing and logistics. With the opportunity to double revenue every 24 months throughout the decade, it is imperative for us to use all possible resources to achieve our full potential.

Moving to Aspen Battery Materials, which we refer to as ABM, we are developing our carbon aerogel technology in the design of low-cost, high-performance, silicon-rich anodes in lithium-ion batteries. The collection of attributes of our carbon aerogel technology creates an ideal protective host or scaffold for silicon and helps address the challenges posed by silicon expansion during each charge cycle. The nearly \$10 million of investments that we are making in scientists, engineers and facilities throughout 2021 are resulting in significant progress towards achieving key technical milestones in the areas of energy density and cycle life.

The design of Aspen's carbon aerogel material allows lithium ions to move more easily across -- sorry, to more easily assess the capacity of silicon while simultaneously preserving the structural integrity of anodes and extending cycle life. The events have also increased our production capacity to deliver larger samples of optimized materials to our partners. This approach to partner engagement plays to the strength of our technology and we believe will enable the broadest and quickest adoption of carbon aerogel solutions in the EV battery market.

When we provide our overall financial outlook for 2022, we will also share specific technical and commercial targets for ABM that will set the course for what we believe will be significant value creation.

At the beginning of this year, we said that our target would be to double revenue from 2021 to 2023 and again from 2023 to 2025. And at the last earnings call, we said that we believe that we have the opportunity to double revenue every 24 months through the decade.

With an accelerating battery performance and safety business led by our PyroThin thermal barriers an important energy infrastructure business driven by resource efficiency, asset resiliency and safety, our rich and commercially leverageable aerogel technology platform and an outstanding team of people to drive the significant scaling of the company, we are confident in our ability to achieve these targets.

In closing, I want to take a moment to recognize Aspen's 20-year anniversary, which we are celebrating this quarter. We have taken our collective vision and executed a strategy that is transforming the company to create the greatest value for our customers and for society as a whole. Even with 20 years behind us, we feel our work and important contribution are just beginning. We, as a company, have a deep gratitude to all of those who have provided wisdom and support to us along the way.

And with that, John, over to you.

John F. Fairbanks

VP, CFO, Principal Accounting Officer & Treasurer

Thanks, Don. I'll start by running through our financial results for the third quarter of 2021 at a summary level.

Total revenue grew 26% to \$30.4 million from \$24.2 million in the third quarter of 2020. Net loss increased to \$7.8 million or \$0.24 per share this year versus a net loss of \$6.8 million or \$0.25 per share in the third quarter last year. I want to highlight that net loss this year reflected a \$3.7 million gain on the extinguishment of debt.

Adjusted EBITDA was negative \$7.8 million this year compared to negative \$3.2 million in the third quarter of 2020. We define adjusted EBITDA as net income or loss before interest, taxes, depreciation, amortization, stock-based compensation expense and other items that we do not believe are indicative of our core operating performance. In the third quarter this year, these other items included the \$3.7 million gain on the extinguishment of debt.

I'll now provide additional details on the components of our results. First, I'll discuss revenue. Total revenue increased by \$6.2 million or 26% to \$30.4 million. This increase in third quarter revenue was principally driven by a continuation of the initial stages of the post-COVID recovery in the global refinery and chemical market, particularly in the United States. And nearly \$1 million of incremental revenue in the EV thermal barrier market, offset in part by a decrease in project work due to the conclusion of the PTT LNG project.

Total shipments for the quarter increased by 32% to 9 million square feet of aerogel blankets, while our average selling price decreased by 4% to \$3.36 per square foot. The decrease in average selling price reflected an increase in the mix of lower-priced 5-millimeter products this year versus the third quarter of 2020.

Next, I'll discuss gross profit. Gross profit was \$3.1 million or 10% during the third quarter this year versus \$1.9 million or 8% during the third quarter of 2020. This increase in gross profit was principally

driven by the 32% increase in shipment volume and a decrease in material costs, offset in part by an increase in manufacturing expense and the 4% decrease in average selling price.

Next, I'll discuss operating expenses. As we've discussed in prior calls, we are strategically increasing our investment in our electric vehicle market opportunities. Our strategic investment this year includes increased spending to enhance the capabilities of our technical, operational and commercial teams supporting our thermal barrier business; the research and development team supporting our carbon aerogel battery material opportunity and our legal resources to expand and defend our IP portfolio.

In contrast, during 2020, we purposely reduced compensation and spending levels throughout the company in response to the uncertainty associated with the COVID-19 pandemic. As a result, third quarter operating expenses increased by \$6 million versus last year to \$14.6 million.

Next, I'll discuss our balance sheet and cash flow for the quarter. Cash used in operations of \$6.1 million reflected our adjusted EBITDA of negative \$7.8 million, offset in part by a \$1.7 million decrease in working capital investment during the quarter. The decrease in working capital investment was broad-based and included decreases in accounts receivable and increases in accounts payable and accrued expense balances.

Capital expenditures during the quarter of \$2.3 million. We're focused on improving the efficiency and reliability of our East Providence plant and the initial engineering work for our second manufacturing facility.

Cash provided by financing activities of \$1.6 million was comprised of \$800,000 generated by our final sales of equity through our ATM facility, and \$800,000 from employee option exercises. As a result, we ended the quarter with \$95.5 million of cash, net current assets of \$98.9 million, no borrowings under our revolving credit facility and shareholders' equity of \$142.5 million. We also had access to an additional \$9.5 million available under our revolving credit facility at quarter end.

I want to highlight that during the quarter, we applied for and received full forgiveness of our \$3.7 million PPP loan from the Small Business Administration.

I'll now turn to our full year 2021 outlook. We expect continued revenue growth this year associated with the initial stages of a post-COVID recovery in the global energy infrastructure market and solid demand growth in the European green building materials market. In addition, we're increasing our outlook for PyroThin thermal barrier revenue by \$5 million to a total of between \$6 million and \$7 million for the year. As Don discussed, this increased thermal barrier outlook is related to an expanded scope for thermal management, mechanical stability and fire protection properties provided by our PyroThin thermal barriers that enhance EV battery performance and safety. In total, as a result, we're increasing our 2021 revenue outlook by more than \$10 million to between \$122 million and \$128 million for the year, equivalent to growth of between 23% and 28% compared to 2020.

We're continuing to invest in people and assets to capitalize on our rapidly growing e-mobility opportunities. This investment includes planned incremental expense to enhance our technical, commercial and operational capabilities and resources in support of our thermal barrier and battery materials opportunities. This investment also includes our planned capital expenditures to design and construct our advanced thermal barrier center, to establish thermal barrier fabrication operations, to expand our carbon aerogel battery materials production, fabrication and testing facilities and to design and construct our second silica aerogel manufacturing plant.

Given the accelerating pace and scale of our PyroThin thermal barrier opportunities, we're increasing the target capacity for our second plant and anticipate we'll be -- we will bring on a greater proportion of the plant assets during the second half of 2023. As a result, we expect to incur a greater proportion of the required capital expenditures in the earlier stages of the project.

Our engineers are focused on designing a greener and more dependable facility with more than twice the capacity of our East Providence plant and lower per unit operating costs. We're also considering establishing fence line production of critical raw materials with a few of our principal suppliers to shorten

our supply chain and reduce our raw material costs. We expect to have the facility operational during the second half of 2023.

We're also planning to establish a fabrication facility likely in Mexico in support of our PyroThin thermal barrier opportunity. We're planning to have the facility established in mid-2022, and fully automated operations in place beginning in early 2023. We believe these strategic investments will position Aspen to take advantage of the significant growth opportunities available to us today in the electric vehicle market and to leverage our aerogel technology platform to develop new high-growth businesses. We'll provide additional detail on these strategic investments and associated financing arrangements as our planning efforts progress during the next few quarters.

Our revised 2021 full year outlook is as follows. We expect total revenue of between \$122 million and \$128 million, net loss of between \$29 million and \$31 million, adjusted EBITDA between negative \$18.5 million and negative \$20.5 million, EPS of between a loss of \$0.96 and a loss of \$1.02 per share. This EPS outlook assumes a weighted average of 30.4 million shares outstanding for the year. In addition, this 2021 outlook assumes depreciation of \$8.9 million, stock-based compensation expense of \$5.1 million, a gain on extinguishment of debt of \$3.7 million and net interest expense of \$200,000. This full year outlook also projects a gross margin in the low to mid-teens and an average selling price of approximately \$3.35 per square foot.

Turning to our outlook for cash. We're currently project capital expenditures of approximately \$15 million for the first -- for the full year. As our planning efforts progress during the next few quarters, we'll provide our investors with multiyear capital expenditure projections to include our second aerogel manufacturing facilities, our Mexican fabrication operations and other capital assets required to support our electric vehicle business opportunities. We will want to raise capital to fund these facilities, equipment and people to keep pace with the expanding scale of our e-mobility opportunities. Importantly, though, we're confident our \$95.5 million cash balance and available credit under our SVB facility will be sufficient to fund our near-term operating and capital expenditure requirements. I'll now turn the call over to Sarah for Q&A.

Question and Answer

Operator

[Operator Instructions] We will now take our first question from the line of Eric Stine with Craig-Hallum.

Eric Andrew Stine

Craig-Hallum Capital Group LLC, Research Division

So thanks for all the detail and good to hear about acceleration of the activity in the revenues in '21 and '22. I'm wondering if you could maybe just go into detail on that a little bit. I know part of that is increased content, but just curious, is that also more volumes with the 2 customers that you have in hand? I mean is that visibility into additional customers? How should we think about that?

Donald R. Young

President, CEO & Director

It is principally content. The number of vehicles is pretty consistent with earlier expectations. But for us, it's been more of a content issue, Eric, as we've been asked to think more broadly about what role aerogels can have more broadly in battery performance and safety. So some things beyond the thermal runaway issue that we've been so focused on in the past year or 2. And these things include broader thermal management challenges that these batteries have and certain mechanical performance that we are addressing as well with our PyroThin thermal barriers. So those things have added up to additional content per vehicle. And that's the reason to a large degree, we're having the kind of revenue that we're projecting and that we're estimating, I should say, in Q4 and that we suggested that we would have 2x or more the revenue that we were anticipating for 2022.

Eric Andrew Stine

Craig-Hallum Capital Group LLC, Research Division

Got it. And is that additional content? Is that with both OEMs? Or is it skewed to one versus the other? I know the one you've had in hand for the longest time, that one is the platform versus the other one being the vehicle, but I'm curious on that.

Donald R. Young

President, CEO & Director

I think what I would like to say, Eric, just from a confidentiality point of view is that this is being applied -- this idea is being applied to several of the companies that we're both serving and that are in our development funnel. So this is a pretty universal opportunity for us as we strive to become this idea of a technology partner. We're very engaged with several of these companies at the design level. And again, it's giving us the opportunity just to address more issues, including issues that take place during the normal course of the battery operation.

Eric Andrew Stine

Craig-Hallum Capital Group LLC, Research Division

Got it. And maybe sticking with the Asian OEM and it sounds like that potentially is announced soon who that is. But just curious where your confidence level lies in winning the entire platform. I know you're starting with this one because they are under some time constraints, but thoughts about getting the rest of it.

Donald R. Young

President, CEO & Director

Yes. We've worked very closely with the company, continue to work very closely. We're delivering production parts into its first vehicle. Again, we think we're in a strong position to be the partner for the battery platform more broadly, which ought to take place here over the course of the coming few quarters.

Eric Andrew Stine

Craig-Hallum Capital Group LLC, Research Division

Got it. And then last one for me. Just you mentioned you've always talked about the 3 levels of engagement and as things continue to trend well there. I mean is it fair to characterize it kind of number of OEMs are pretty similar in the same buckets just that those -- that level of engagement has just gotten closer to resolution, whether it's an award or what next steps are?

Donald R. Young

President, CEO & Director

I think the level of engagement, the way I would say is that in some sense, it's become more formal. As I said in my comments, we are seeing many more -- we're engaged with many more requests for information, requests for quotation and these kinds of things. So I would just -- it's advanced in the sense that it's more formal in that way, Eric.

Operator

The next question comes from the line of Colin Rusch with Oppenheimer.

Colin William Rusch

Oppenheimer & Co. Inc., Research Division

So as you look at that pipeline, I'm just wondering if we can get an overall scope and scale of what that pipeline looks like from a square meter perspective or revenue number or total customers? And then the second question within that is the customers that you've -- it looks like you're working with are reacting to problems that they've had out in the field. I'm wondering about earlier stage engagement around proactively avoiding some issues around using material in a more structural way potentially to support their programs potentially in other parts of the vehicle beyond just the battery.

Donald R. Young

President, CEO & Director

Thanks, Colin. So on the -- while this is really a pretty fast-moving situation for us to really give you longer-term projections because several of these things are somewhat binary, if you will. And each one can have an enormous impact on our revenue out in 2023 and 2025. And so -- but a lot of these decisions are going to be made here over the coming handful of quarters at the very, very most. And so it's a large opportunity. And in our investor deck, we had some projections in there, and we're very confident in the size that we outlined at that point in time, this -- so it's definitely a multibillion dollar opportunity for us. And again, we're not really projecting much beyond that at the moment. But with each one that we win, it brings a fairly large chunk of revenue our way that is usually defined in a long-term contract.

In terms of your second question, what I would say is that we're principally focused on the battery platforms themselves. I think there is additional work for us around the battery platforms. That is interesting to us, and we have discussions around. But I wouldn't really extend the scope beyond the battery platforms themselves at this point in time. I mean, historically, we have had sales, let's call them transportation sales, sometimes in vehicles, sometimes in -- excuse me, and trains, there are a variety of different types of transportation vehicles. But again, our principal focus and the real drivers of our revenue are going to be our work around the battery platforms themselves.

Colin William Rusch

Oppenheimer & Co. Inc., Research Division

Awesome. And then obviously, there's a lot going on with basic materials from a cost perspective, logistics perspective. I guess just since the end of the quarter, starting with October, can you speak to any adjustments that are happening or concerns that you're seeing in terms of availability of material cost structure, things like that and your ability to pass those costs on to customers?

Donald R. Young

President, CEO & Director

Very topical. We're not different than a lot of companies right now. We have our supply chain and it's stretched frankly, around the globe, and our customers are around the globe. So there are a lot of challenges associated with it. What I would just say about it is in our energy infrastructure business, for example, we have in the past, and it has been largely in the industry practice to be able to pass along these kinds of increased expenses or call them inflationary type expenses increases. And so I think I would expect that we will do the same in this particular case, especially now that it seems quite apparent that it will extend into 2022. And so I think we're in good shape from that point of view with pricing power, if you will. Again, we're not trying to take advantage of the situation, we're just trying to hold serve basically.

Operator

The next question comes from the line of Alex Potter with Piper Sandler.

Alexander Eugene Potter

Piper Sandler & Co., Research Division

Great. Maybe one, I guess, follow-on question to what Colin was just asking there. Less on the supply chain for what you're making now and more on the supply chain for Plant #2. Are there any particularly long lead time pieces of equipment or anything like that, that could be slowed down as a result of the pandemic or these logistical problems? Does that keep you up nights wondering whether you're going to have to delay the project as a result of any of that?

Donald R. Young

President, CEO & Director

Yes. We are very engaged with our -- with the vendors for our long lead time items. And we are in good shape right now with that, Alex. We are not expecting that to be an issue for the startup of Plant 2 by the second half of 2023. I think our challenges are more just around the more current day logistics of moving materials around both our raw materials and our products is just a challenge. It's keeping us up at night, in particular, our supply chain team.

Alexander Eugene Potter

Piper Sandler & Co., Research Division

Okay. Great. And then obviously, this is -- these comments around the expanded scope of work using PyroThin for other purposes that all sounds great, upward revision to the revenue guidance, that all sounds great. I don't know the extent to which you're comfortable talking maybe in slightly more specific terms around what exactly it is that you'll be using PyroThin for. You mentioned sort of general thermal management or some structural, I guess, characteristics of PyroThin. But can you elaborate on that exactly other than just kind of in casing cells or packs or modules to prevent thermal runaway?

Donald R. Young

President, CEO & Director

Yes. Let me give maybe an example of that, that would be helpful. When you think about mechanical performance and mechanical durability, you know that typically in a module that the cells have material between them to keep them in place and a constant pressure on the cell wall. And so as the material -- as the cells are expanding a contraction with each charge cycle. And so if you think about our material, it has a -- it's compressible. Think of it as kind of a blanket-like material. And so there are -- there can be roles for our material, sometimes with other materials as well to play a role in that. And that relates to the normal course of a battery's life as opposed to the extreme situation of thermal runaway, which one hopes does never happen, but we know it does. So that's the kind of thing that we're exploring today.

Alexander Eugene Potter

Piper Sandler & Co., Research Division

Interesting. Okay. That's very helpful. And then maybe last question on the -- maybe the types of companies that are in your funnel as well as your own capacity, I guess, is there a natural limit to the number of inbound inquiries that you can entertain? And as a result, are you maybe rank ordering these inbound requests in order of, I guess, long-term revenue potential or your own perception of the quality

of the customer? I guess maybe that's 2 questions buried in there. Are you dealing with all kinds of customers and treating them all more or less equal? And then second, is there a natural limit to the number of people that you can be dealing with?

Donald R. Young
President, CEO & Director

Yes. So in terms of ranking, look, everyone is important to us, but some people -- some companies are probably more important than others in some sense because of scale, because of the likelihood of them being successful, the breadth of their programs, the technical prowess of their teams. Those sorts of things really draw us in and we engage very well with those kinds of groups. So I think there is some natural prioritization going on as you would think. We are interested in a variety of areas, including stationary energy storage and some other areas of aviation and what have you. But just from a breadth point of view, it's hard to compete with one of the 3 or 4 or 5 largest automotive OEMs in the world or multiple of them. So -- and they can really absorb a lot of our efforts.

In terms of sort of the how many can you sort of take on? And if I understand that part of your question correctly, I think what I would say is -- and John said it in his comments and I said it in my comments, we are looking very carefully at the demand profile and that translates for us to the way we build Plant 2. And we've talked -- we've described when we built Plant 1. We did it in 3 phases, basically. And our initial thoughts about Plant 2 were to do it in a very similar kind of way, bringing on the capacity sort of incrementally as you lock down business. And as this business comes out as more rapidly and we're having significant success within that development funnel and our confidence in progressing through the RFI, the RFQ processes increases. We are looking at building more of that plant during Phase 1, let me just say. And -- but it's going to be demand-driven. We're not building and hope they come. It's going to be demand-driven, and we're working real hard to sort of coordinate those things.

And as I said, the business sort of comes in chunks. And so at least the contracts sort of come in chunks and then they roll out their vehicles. And we all know, Alex, you more than most, some of them are going to be more successful than they're suspecting and some are going to be less successful than they're suspecting. And so that's why our strategy around being OEM agnostic is really important for us. And our intent to be the industry standard here as -- in our PyroThin thermal barriers is just really critical to us. So I hope that sort of answers your multi-part question, but that's the way we're thinking about the business.

Operator

The next question comes from the line of Jed Dorsheimer with Canaccord Genuity.

Jonathan Edward Dorsheimer
Canaccord Genuity Corp., Research Division

Thanks for all the details and nice job on the quarter. First question for you. I guess the Georgia -- the second facility down in Georgia, that is Statesboro, correct? That hasn't changed?

Donald R. Young
President, CEO & Director

We're referring to it as Southeastern United States at the moment.

Jonathan Edward Dorsheimer
Canaccord Genuity Corp., Research Division

Okay. Are you in -- but in the Q, you've previously announced or in your case, the breaking ground of a facility in Georgia. Is it -- I guess, is it for negotiations with -- between municipalities for -- is that sort of the stage that you're at in terms of where the location and what incentives may be?

John F. Fairbanks
VP, CFO, Principal Accounting Officer & Treasurer

Jed, I can clarify. So back in 2016, we had booked through Georgia and South Carolina. And at that time, we had selected Statesboro, Georgia for the location of a second facility. But we've reconsidered. We're looking at multiple locations, and we have selected a site now, but we haven't -- we have not announced that it is Statesboro, Georgia. Georgia or South Carolina, we've just said Southeastern in the United States.

Jonathan Edward Dorsheimer

Canaccord Genuity Corp., Research Division

Got it. That's really helpful. And I'm assuming when you do announce -- yes, go ahead.

Donald R. Young

President, CEO & Director

No, no, I was just going to say I think it's where you're headed. We will announce jointly with the Governor's office and we'll do this properly, let's say, in the not-too-distant future.

Jonathan Edward Dorsheimer

Canaccord Genuity Corp., Research Division

Understand. Great. And good luck with the incentives for that, too. On the Mexico facility, when you say the production, one, I guess, have you targeted a location? It sounds like you're further along there. So I'm curious if you have settled on a location there. And then I just have -- just curious about some of the details, square footage or how we should think about production output from that facility?

Donald R. Young

President, CEO & Director

Yes. So what we're doing in Mexico is, as we've described our PyroThin thermal barrier, it's a multipart system. And the automated fabrication of those parts that can equal millions of parts we're doing in Mexico. We have had a team. They're looking at different sites. We do have an adviser in country that is guiding us. We've also leaned on the resources that [coke] industries have, and they have substantial fabrication facilities in Mexico. So we're trying to learn all the lessons and get as smart as we can about this, and we are making good progress, and we would expect to have an operation there by the end of 2022. So it's -- we're moving right along with the process that we have not quite picked a location yet. In terms of size, John, do you have -- I don't have that really with me.

John F. Fairbanks

VP, CFO, Principal Accounting Officer & Treasurer

We're not -- it's all part of our planning at present. So Jed, once -- it has the same way we're sizing our aerogel plant in the Southeastern U.S. We're also engaged in an effort to size the fabrication facility. I think the important part is we'll pick a lease -- we'll lease a facility. So we're not going to have an extended lead time to get into a building. And it's much less capital intensive and the equipment that we would actually put into that plant is more readily available. And so the lead time to get into a facility is measured in months and the lead time to have automated fabrication operations in place is now maybe a year. And it will absolutely meet our needs and allow us to keep pace with demand for our thermal barriers.

Jonathan Edward Dorsheimer

Canaccord Genuity Corp., Research Division

Got it. Maybe just shifting gears. I didn't hear any update on the silicon anode. And just curious how that project is developing? Any comments there?

Donald R. Young

President, CEO & Director

Yes, thanks. No, I did mention that we will have invested over the course of 2021, roughly \$10 million this year in scientists and engineers and in facilities. And what it's allowed us to do is produce larger quantities of material, better materials and deepen our engagement with our announced partners and other partners

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as well or other entities we're working with. And so our focus, Jed, remains on really leveraging the unique aspect of our carbon aerogel in the protection of the silicon through its life cycle. And so we're making very good progress. We're iterating faster with these other companies.

And we just didn't have enough material before we made these investments to really keep pace and work at a large enough scale to satisfy them, and we're now doing that. And it's been -- this has been true for just about one full quarter now. So the pace has quickened. And we're really excited about the work. And what we've committed to is that when we provide our 2022 guidance for the year, we will add at that point in time some both commercial and technical milestones for acid battery materials to help you guys keep score just a little bit better throughout the year 2022.

Operator

The next question comes from the line of Amit Dayal with H.C. Wainwright.

Amit Dayal

H.C. Wainwright & Co, LLC, Research Division

Just really quickly on the expectations for next year from thermal runaway of PyroThin, you're doubling revenue expectations. Is this coming from visibility from customers you have in hand? Or are you expecting additional orders from other customers that you are working with?

Donald R. Young

President, CEO & Director

So this is really coming from additional content from the customers we have, Amit. and so that confidence was not coming -- was not necessarily derived from winning more contracts and delivering production parts in 2022 from other entities that -- think of that as upside. Our comments that John and I provided really related to content per vehicle from the 2 major automotive OEMs that we have one, U.S.-based, one Asia-based.

Amit Dayal

H.C. Wainwright & Co, LLC, Research Division

So if you do get additional orders from other customers, will you potentially be able to meet that demand with the capacity you have in hand?

Donald R. Young

President, CEO & Director

Yes, we do. We have capacity today to reach up into the mid-200s millions of dollars. And so John's guidance was in the [\$118 million to \$128 million] for this year. And so we've got some room to run here before we run out of capacity. And we believe that we can double revenue from 2021 to 2023 with the capacity that we have today. So we've got room to run. But we are, as we said, expediting our work on Plant 2.

Amit Dayal

H.C. Wainwright & Co, LLC, Research Division

Yes. So similar question I had was you don't necessarily have to give up any energy infrastructure business to capture the EV opportunity fully for at least 2022?

Donald R. Young

President, CEO & Director

That's correct. And look, our energy infrastructure business is very important to us. And we play an important role, I think, in that industry. Our distributors, our contractors, our end users rely on this product, and we believe that we can continue to grow that business as well. Even if you subscribe to the idea that traditional energy might go sideways for the next 3 decades. We believe that just given our resource efficiency, asset resiliency and the safety nature of our products, that those attributes will become at least or only more important to these companies as they try to achieve some of their own

sustainability goals here over the course of the coming 5 and 10 years. So -- and so we do have room, and we are anticipating that we'll continue to grow that business.

Operator

The next question comes from the line of Chris Souther with B. Riley.

Christopher Curran Souther

B. Riley Securities, Inc., Research Division

Just a quick one on the second facility planning. We're talking about more than 2x the size of the current facility. But are we going to get a full picture of the size, scope and costs like with the governor announcement you think? Or is that something that comes later or next year or something?

Donald R. Young

President, CEO & Director

Yes. Look, I -- go ahead, John. Yes.

John F. Fairbanks

VP, CFO, Principal Accounting Officer & Treasurer

Yes. We know that we need to give that to you. It will definitely come by the time of the governor -- the announcement with the governor. And the way we think about it internally is we want to tell you what the return is on the investment when we talk about the investment. And so we'll make sure you understand the scope, the scale, the capacity, the operating efficiency of that plant. And we're targeting to have that out in public within the next few months.

Christopher Curran Souther

B. Riley Securities, Inc., Research Division

Okay. That's great. And then maybe just the CapEx cadence that we should be looking at. And I think the third quarter will be in the Q there. But fourth quarter, are we kind of starting to order any of the long lead?

John F. Fairbanks

VP, CFO, Principal Accounting Officer & Treasurer

Yes. We're starting to put in orders for long lead time equipment, although most of that would -- that cash would actually flow and it should be recognized as capital expenditures in 2022. We did give the guidance of \$15 million in CapEx for the full year 2021.

Christopher Curran Souther

B. Riley Securities, Inc., Research Division

Got it. Okay. And then you kind of brushed on potential new wins for next year upside. I just wanted to get a sense when was essentially just a few months before the vehicle launch. So what vehicle launch time line for some of the Stage 3 kind of late-stage folks? Is it -- they're kind of looking at vehicles potentially for 2022 launches? Or is it more a '23 and beyond after you have more [capacity] there?

Donald R. Young

President, CEO & Director

Yes. Chris, I think that wins that we will get during 2022 and even the first half of 2022 will really begin to materialize in terms of revenue in the following year and then typically really kick in much more substantially in 2024. That's the pattern that we've seen in the contracts that we've won. And it's consistent with what we all read about projections and frankly, the number of launches that these 7 or 8, 9, 10 companies have scheduled between now and even as late as 2025. 2025 models typically in 2024. So look, I think the majority of the revenue that we have in 2022 will come from the 2 customers we have today. And -- but switching over to production parts with some of these entities in the latter part of 2022. Again, really sort of sets the stage for a nice acceleration in 2023.

Christopher Curran Souther

B. Riley Securities, Inc., Research Division

Yes. So on 2023, you're talking about launching the second facility here in the back half. Just do we have a sense of size of particular modules at that facility? We get a sense of kind of the cadence of that over \$500 million coming online. What would be kind of the time frame we should think about, if you don't want to say how much bigger than \$500 million? How many stages and when we should expect kind of the different lines to come on?

John F. Fairbanks

VP, CFO, Principal Accounting Officer & Treasurer

Yes. We're going to -- that's exactly the kind of detail that we hope to get out within the next few months and no later than in our Q4 earnings call. We'll provide the overall size of the facility, what the CapEx costs are, when we expect to incur those costs. Ultimately, what the capacity of that facility will be to keep pace with revenue. We obviously give you our 2022 guidance at that point in time as well. And we'll talk about how it's phasing in. So we want to give you the whole picture at once so that you can understand the dynamics of the business and not just the investment cost, but the return on the investment and what we're seeing moving forward.

Donald R. Young

President, CEO & Director

Chris, I think I would add -- the thing I would add to John's comment is it's -- I think they call it kind of the Goldilocks thing, right, we don't want too big to small. We certainly don't want to build this thing too small so that we're out of capacity before we get started because this is the time to win these contracts, and this is the time to deliver. And these are longer-term contracts. And so you've got to be capable. And so that's our focus here. And again, as John says, we're really trying to match up our demand profile with the size of that first phase. And of course, we're leaning towards building a bigger in the beginning because that's what we're getting from the market.

John F. Fairbanks

VP, CFO, Principal Accounting Officer & Treasurer

Yes, the opportunity is coming. It's larger and it's coming at an accelerated pace, above even what we expected a quarter ago.

Operator

The next question comes from the line of Tom Curran with Seaport Research Partners.

Thomas Patrick Curran

Seaport Research Partners

Guys, thanks for squeezing me in, even though we're over the hour now. I'll be quick. I've only got 2 left. Don, at this juncture, does the apparent front runner to become customer #3 for thermal barriers seem most likely to extend you a design award for a single EV model or to ink a multi-model platform-wide contract, similar to customer #1. And what's the earliest you might secure that third customer? Could we see an announcement on that potentially before the 4Q earnings call at the soonest?

Donald R. Young

President, CEO & Director

So it will be more battery platform in nature. So I mean I'm speculating here a little bit. There's a roster or there are multiple OEMs in this RFI, RFQ product validation activity level today. I'm anticipating that the most sort of notable ones are more platform-oriented and I'm anticipating that.

So our earnings announcement is late February. So that's a pretty short period of time. So I don't want to set expectations in that way, but I would just say that over the course of the next couple of quarters, we'll continue to talk about the funnel and make announcements as we win them. There are going to be some

decisions made though in this kind of 2 quarter time frame. And we think we're in a strong position, again, with some of the very largest companies and with some of the newcomers as well.

Thomas Patrick Curran
Seaport Research Partners

Great. We're eagerly staying tuned. John, when it comes to Plant #2, could you just revisit that very helpful heuristic you've previously provided when it comes to the expected CapEx outlay per dollar of incremental revenue you'd be adding be an investment in additional production capacity? Does the \$0.65 of CapEx per dollar of incremental revenue in production capacity, does that still hold? Or based on the work you've done thus far, would you update that for Phase 1 of Plant #2? And then when it comes to Phase II in the second half of 2023, do you expect that just to be a seamless transition? Or might we see a pause between phases?

John F. Fairbanks
VP, CFO, Principal Accounting Officer & Treasurer

So yes, I'll answer your second question first. We could potentially see a pause between phases. I think the real question because we talk about revenue capacity coming out of the plant. We had metrics that were for traditional aerogel products alone, but now we've got these multipart thermal barriers that add additional content, additional revenue. And so I'm going to withhold making a comment on that at present until it really is going to be dependent upon the revenue that we get for the thermal barriers with aerogel just being a component of that overall revenue. I would say, however, there is inflation out there, steel costs are up, a big part of the cost of building this plant is steel. And then construction resources are in high demand. And just like every other business out there construction companies are having to pay up in order to get workers. So inflation is working against us.

The additional content on our thermal barriers is working for us. And we'll put it all together and when we talk about the investment in the asset, we'll make it very clear what's the capacity, what's the aerogel capacity, what's the aggregate thermal barrier revenue capacity that comes from the investment we make. And so we'll make sure we package that and provide all that detail when we talk about the plant more fully in the next few months.

Operator

The next question comes from the line of Chip Moore with EF Hutton.

Chip Moore

Just quickly on the energy infrastructure side, guys, I don't think we touched on LNG. Maybe you can update us on the pipeline there and then anything on the subsea side in terms of trends?

Donald R. Young
President, CEO & Director

Well, on the -- I think we've all been reading about the importance of LNG in Asia and Europe and as a production center here in the United States. So I think activity levels are high. The pandemic did cause project work in general to slow down. And we've talked about that a lot. Frankly, it caused maintenance work to slow down as well. We're seeing maintenance work come back quite nicely, some pent-up demand for sure. Project work is -- I think people are still a little hesitant to pull the trigger on a major project that has insecurity around workforce and the continuity of executing on that project. So having said that, we are in the specifications of a number of projects, and we're staying close to those projects. And we think, again, we'll win our fair share of them. Both LNG and chemical refinery type projects as well. So we're -- we feel really strongly about our energy infrastructure business and our place in that business in the years to come.

The subsea work has always been a business I think dating back for most of our 20 years. It's been a business that typically has been somewhere between \$5 million and \$15 million in size. It's been sort of an important solid sort of niche for us. And I think it will continue to be here in 2022. And when we look out in our pipeline, we sort of keep about a 3-year pipeline of sort of proposed projects subsea. Again,

there is a nice handful of projects in 2022 that could materialize, and I would expect this to come fall into that \$5 million to \$15 million kind of range again in 2022. So again, just relatively consistent.

Chip Moore

Yes. No, that's helpful, Don. And obviously, we're seeing a number of things every day of high-profile announcements in terms of fill out of battery supply chains here in the U.S., Canada, Europe. Wondering maybe if you could speak a bit to implications for your opportunity set maybe from a geographic perspective as well as some of the more advanced battery material opportunities, even though they're a bit further out on the horizon.

Donald R. Young

President, CEO & Director

Yes. Yes. No. Look, we're very engaged with these companies. We mostly talk about automotive OEMs. But we're -- when you talk about thermal barriers. But we're also engaged with the Tier 1s, with the battery OEMs themselves. And then you switch over to the Aspen battery materials side, our carbon aerogel. And again, we have the belief on the carbon aerogel side that we're going to see, sort of -- I guess, I would say, sort of consortiums of companies come together and play important roles in sort of next-generation batteries. And our goal is to be sure that our carbon aerogels join in on some of those consortiums. And we -- and those will be comprised of large battery OEMs, and some of the newcomers on the battery OEM side. But we also know that the automotive OEMs have significant battery programs, battery chemistry programs today.

So I think you'll see, let's call it, a materials company like ours, pairing up with a battery OEM. Big and small, pairing up with some of the large automotive OEMs. So I think you're going to see that play out here. And again, we believe that our carbon aerogels are special, and we'll find some daylight in those activities.

So that's what we're working towards. Again, I don't want to convey that this is a 2022 or 2023 revenue item for us. But we do think that it can be value generating as we get further third-party validation for our work, and we begin to be able to compare our data with some of the other companies that have worked hard in the battery material space as well.

Operator

The next question comes from the line of Jeff Osborne with Cowen.

Jeffrey David Osborne

Cowen and Company, LLC, Research Division

Just had a couple of quick ones. I think in the past, you had talked about \$100 to \$300 content per vehicle. It sounds like that might be a bit higher now. Can you give us an update on where that should be?

Donald R. Young

President, CEO & Director

Yes. It is a valid number for some cases. And I think as -- and I think we will be at the higher end of that range or extend beyond that range with some of the other cases where, again, we're playing a broader role in the battery construction, if you will, the battery performance and safety, where we're addressing some of the challenges around a battery module during its normal course of operation as opposed to the mission-critical part around thermal runaway. So when we get into those situations, we're expecting a content per vehicle to be greater than that \$100 to \$300 level. And it is the principal reason why you're seeing us increase expectations for 2021 and 2022.

Jeffrey David Osborne

Cowen and Company, LLC, Research Division

Got it. That's helpful. And is it both the North American and Asian customer that it's expanded the scope or just one of the 2?

Donald R. Young

President, CEO & Director

Again, we rather -- we'd rather maybe not say. I mean if there were 10 of them, I would say, well, 3 of them, but it's -- this is going to get revealed here pretty soon who these companies are, and I just would like to maybe...

Jeffrey David Osborne

Cowen and Company, LLC, Research Division

Understand. Maybe just 2 other quick ones given out of interest of time, one for Don, one for John. Don, for you, on the Asian customer, can you -- you mentioned that there's like a secondary party that might be using the platform. Is the award that you have just for one vehicle now, but there's a desire or an ambition potentially to expand that? Or is it literally just a one model OEM?

Donald R. Young

President, CEO & Director

Yes. No. With our Asia-based win, it is for a specific model that will be marketed in Asia, in Europe and in North America. But most of our engagement with the company has been -- has included the team that's responsible -- a team -- one team that's responsible for the vehicle and another team that's responsible for the battery platform overall. And so they made commitments to launch that vehicle before their platform was completely designed. And so we're engaged with both of those teams. We've obviously won with the vehicle. And we're in -- again, we believe we're in excellent shape to win the platform longer term.

Jeffrey David Osborne

Cowen and Company, LLC, Research Division

Got it. That's helpful. And then the last question I had is, John, you mentioned the \$10 million of incremental investment in '21. A common question we get is sort of the path to profitability. And obviously, there's a lot of incremental margins that flow through as a result of utilization, but it's a bit unclear how much OpEx you guys need to invest in all of these additional initiatives. So I know you're not giving 2022 guidance, but would you say that \$10 million is a drop in the bucket that of what's needed as it relates to, say, '22 and '23? And there's a pretty sharp hockey stick of incremental investment for each of these new initiatives? Or do you expect it to be substantive or flattish at these levels?

John F. Fairbanks

VP, CFO, Principal Accounting Officer & Treasurer

Yes, the general profile, I think, makes sense. We've said single-digit millions next year from our initial 2 customers. We just updated that 2x, which would put it somewhere between \$15 million and \$20 million in that range. So when you think about a \$10 million operating expense investment or something even more significant than that, the EV business is going to generate losses in the 2022 time frame, but you get a very significant expansion of revenue potential in '23 and leading into '24 and '25. So it definitely -- we will see losses coming from the EV business in 2022, but then sharp increases in profit contribution from that business and resulting ultimately in profitability beginning in 2023 and clearly into 2024. So that's the general profile. We'll be much more specific, though, when we talk about the sizing of the plant when we get 2022 guidance. We'll also try to give you a really good sense for the revenue ramp and the path to profitability at that time as well.

Operator

At this time, there are no additional questions. So I will pass it back to the management team to provide closing remarks.

Donald R. Young

President, CEO & Director

Thank you, Sarah. Appreciate it. Hey, we appreciate your interest in Aspen Aerogels, and look forward to reporting to you our fourth quarter 2021 results in late February. Be well, and have a good evening. Thanks so much.

Operator

That concludes the Aspen Aerogels Q3 2021 Earnings Call. Thank you for your participation, and enjoy the rest of your day.

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